

# Final Report: SARIMA Modelling of Singapore's Total Fertility Rate and Total Live Births

## Abstract

Total Fertility Rate (TFR) and Total Live Births (TLB) for Singapore spanning 1960 to 2025 were analysed and determined if a Seasonal ARIMA (SARIMA) model can completely capture the long term decline. Additionally if it could capture the 12 year cyclical nature of the data, given contextual background on Chinese Astrology with relation to Zodiac Animals.

It's seen in **Figures 1** and **Figures 2** that this data exhibits strong downward trends and a non-constant variance with visible peaks and troughs in Dragon and Tigers years most prominent from 1970s onwards. Supporting the notion of a 12 year cycle.

With TFR, the selected model  $SARIMA(4, 1, 0)(1, 1, 0)_{12}$  has the lowest forecast error in **Table 5** and is consistent with white noise assumptions presented in **Figure 5** and **Table 6**. This model has captured the declining trend and remained within 95% expectation of a Time Series shown in **Figure 9**. However, has failed to fully capture the 12-year cyclical pattern in Be it due to lack of data or a more complex modelling is required.

TLB, it's best possible model was of the same structure as TFR, yielding the lowest root mean square error among other potential models in **Table 9**. Further residual diagnostics presented in **Figures 8** and **Table 10** supports this. However, forecasts in **Figure 10** shows high uncertainty and insufficiently capture the decline in live births or the cyclical nature.

SARIMA models were successful in modelling the declining trend in total fertility rates in Singapore, producing forecasts for the this variable that were within expectation despite not capturing the 12 year cycle effectively for father into the forecasts. Modelling for total live births was however far less fruitful in capturing the more volatile and cyclical nature of the data. Suggesting that SARIMA does explains the data well enough, additional resources, more complex modelling may be required to model Zodiacal lead family planning.

## Introduction

The dataset has been provided by the Singapore Department of Statistics<sup>1</sup>. It primarily covers births and fertility rates within Singapore, collecting data concerning a variety of variables from 1960 to 2025. However for the purpose of this data analysis and modelling the main variables that will be considered are<sup>2</sup>:

- Year:
  - The EDA will be concerned with the entire period of 1960 to 2025. However for modelling and forecasting, it pertains to the periods 1960 to 2012, and 2013 to 2024 respectively.
- TLF: Total Fertility Rate
  - This refers to the average number of live-births each female would have during her reproductive years, subjected to prevailing age-specific fertility rates in the population in the given year.
  - Additionally data prior to 1980 pertains to total population while onwards pertains to resident population (Singaporean citizens and permanent residents).
- TLB: Total Live Births
  - Data is concerned with all live births within Singapore and its territorial waters as registered by the Singaporean government at the year of birth.

Singapore's demographic composition is majority ethnically Chinese<sup>3</sup>. This is worth noting as Chinese astrology, the culture and traditions relating to such have a sizable effect on the dataset provided. To further elaborate, aside from the typical Gregorian calendar used for dating. A lunar based calendar corresponding with the zodiac cycle is used to mark Lunar New Year for populations of Chinese descent or cultural connection, such as China, Taiwan, Hong Kong, and especially so for Singapore. Additionally, in Chinese astrology there is a belief that an individual's personality, characteristics and future prospects are based on what zodiac animal the corresponding Gregorian year is<sup>4</sup>. According to *Priceconomics (2015)*, for Singapore, there was conscious effort by parents to time pregnancies so they align with the year of the Dragon. This is mainly because dragons are often associated with qualities of honesty,

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<sup>1</sup>Singapore Department of Statistics. (2026). *Births and fertility rates, annual* [Data set]. data.gov.sg, [https://data.gov.sg/datasets/d\\_e39eeaeadb571c0d0725ef1eec48d166/view](https://data.gov.sg/datasets/d_e39eeaeadb571c0d0725ef1eec48d166/view)

<sup>2</sup>Singapore Department of Statistics. (2026). *Births, crude birth rate (CBR), total fertility rate (TFR) and reproduction rates* [Data set]. SingStat Table Builder, <https://tablebuilder.singstat.gov.sg/table/TS/M810091>

<sup>3</sup>Cultural Atlas n.d., *Singaporean culture: population statistics*, SBS, <https://culturalatlas.sbs.com.au/singaporean-culture/singaporean-culture-population-statistics>

<sup>4</sup>Priceconomics 2015, *How the Chinese zodiac affects national birth rates*, Priceconomics, <https://priceconomics.com/how-the-chinese-zodiac-affects-national-birth/>

magnanimity and power. Conversely there is an aversion to years corresponding with the Tiger, as they're often associated with disobedience and long term difficulties for themselves and surrounding family. The complete cycle of Zodiac Animals and the most recent years that they correspond with is seen in **Table 1**.

Table 1: The last twelve years and their corresponding Zodiac Animal

| Year | Zodiac  |
|------|---------|
| 2014 | Horse   |
| 2015 | Goat    |
| 2016 | Monkey  |
| 2017 | Rooster |
| 2018 | Dog     |
| 2019 | Pig     |
| 2020 | Rat     |
| 2021 | Ox      |
| 2022 | Tiger   |
| 2023 | Rabbit  |
| 2024 | Dragon  |
| 2025 | Snake   |

Hence, for this TFR and TLB there should be an expectation for peaks and troughs corresponding with Dragons and Tigers respectively. A cyclical pattern should be expected along with a downward trend<sup>5</sup>.

Further exploratory analysis in the data shows that there is a case of a cyclical pattern, a decreasing trend, and peaks and troughs corresponding with Dragons and Tigers seen in **Figure 1** for TFR and **Figure 2** for TLB. It's seen here that this 12 year cyclical pattern becomes more prevalent for Dragon and Tiger years starting from 1976 despite the decreasing trend of fertility rate and live births hence smaller family sizes. Supporting *Priceconomics (2015)* which stipulated that this as a result of smaller family sizes Zodiacal superstitions have influenced family planning. This phenomena has only become more prevalent from the 1970s as Singapore modernised and industrialised.

Therefore, from this exploratory data analysis, given the contextual background of this dataset. *How well can a seasonal ARIMA model explain and capture this 12 year cyclical pattern and forecast for the period spanning 2013 to 2024?*

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<sup>5</sup>CNA 2023, *News analysis: Singapore's historic low fertility rate not worrying yet, with clearer sign to come for 2024 Year of the Dragon*, CNA, viewed 11 June 2026, <https://www.channelnewsasia.com/singapore/news-analysis-dip-singapore-2022-low-birth-rate-2024-dragon-year-5591231>

## Discussion and Conclusion

A limitation is that SARIMA is univariate, it uses past values of the series to forecast future outcomes. Hence, it cannot account for behavioural, cultural or policy related influences especially for timing birth preferences according to *Priceconomics (2015)*. SARIMA isn't designed to account for human behaviour such as this so issues have arisen from modelling the 12-year Zodiac cycle, such as unusually high RSME in **Table 9**.

Additionally, the data from 1970s onwards is more prominent than previous years. Due to modernisation, and industrialisation of Singapore, families as a result have gotten smaller and thus more careful planning around Zodiac cycles<sup>6</sup>. Another limitation would be using exclusively the Total fertility rate or live births in general, when trying to model this zodiac cycle. Despite Singapore being a majority ethnically Chinese there is still a sizable percentage of other ethnic groups that do not typically follow these traditions. So accounting for these, grouping by ethnicity may help clear up the data for better modelling. Furthermore, as the Lunar calendar does not align 1:1 with a Gregorian calendar, there's reasonable oddities with trying to model a seasonal cycle that is culturally specific.

So to conclude, SARIMA models provide a reasonable statistical explanation of Singapore's fertility data. However there are limitations and these findings cannot explain and capture well enough the cultural demographic pattern. The models were able to generate forecasts and capture broadly the data, but not fully capture a cultural phenomena that is present with in the data.

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<sup>6</sup>Business Insider 2024, *Chinese parents of dragon babies: experts on drawbacks in Singapore, competition, education, personality*, Business Insider, <https://www.businessinsider.com/chinese-parents-dragon-babies-experts-drawbacks-singapore-competition-education-personality-2024-8?>

## Methods

Exploratory data analysis, modelling and model validation was done via R version 4.5.1 and RStudio using the package fable package `fpp3`. This package has the following built in:

- `tibble`
- `dplyr`
- `tidyr`
- `lubridate`
- `ggplot2`
- `tsibble`
- `tsibbledata`
- `ggtime`
- `feasts`
- `fable`

Additionally the packages `knitr` and `ggsci` are used for in the Quarto Markdown Document for formatting and presentation. The `.R` files for the EDA, modelling and the `.qmd` for the report along he `.csv` for both the raw and cleaned data are provided in the the github repository. The link to which is provided in the **Statistical Appendix**.

The TFR and TLB for the full data spanning from 1960 to 2025 in **Figure 1** and **Figure 2** shows a decreasing trend. Additionally the variance is not constant over time. For TFR there is higher variability between for example the periods 1960 to 1975 versus lower variability of 1995 to 2024. This is also the case but more prevalent in total live births. Hence, there is heteroscedasticity in both time plots. Additionally, given the contextual background, it is then concluded that there a 12 year seasonal or cyclical pattern present in both variables.

A new data column calculating and aligning the years to which zodiac animals correspond it corresponds to was added to the the dataset. This is to highlight the Dragon and Tiger years regardless of what their TFR and TLB was. Therefore it's seen clearly that from the 1970s, there are noticable peaks of troughs for Dragon and Tiger years. The data is concluded to be not stationary has it possess all three characteristics that violate this assumption being, a trend, heteroscedasticity and a seasonality.

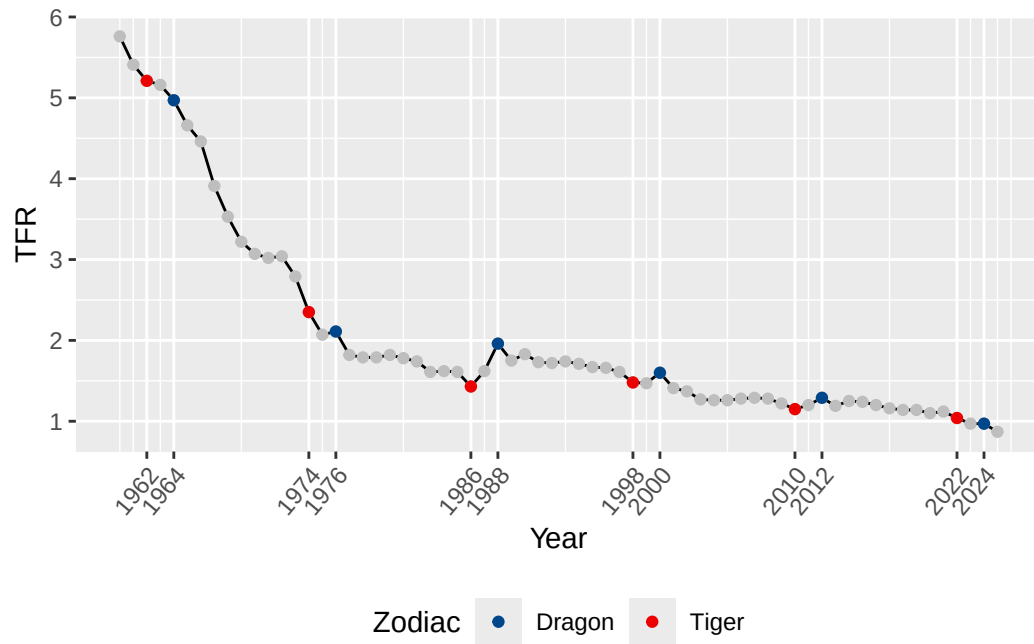


Figure 1: Time plot of the annual Total Fertility Rate with peaks and troughs for Dragon and Tiger years highlighted.

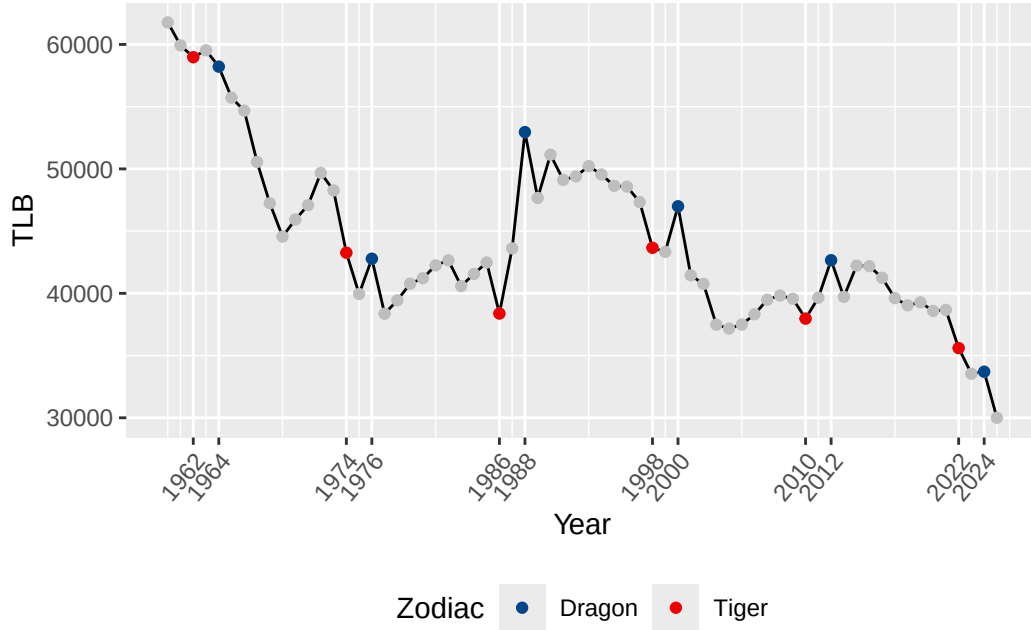


Figure 2: Time plot of the annual Total Live Births with peaks and troughs of Dragon and Tiger years highlighted

Before any transformations and differencing was done, the data was split into two sets, A training set spanning the periods from 1960 to 2012, and a testing set that only contains 2013 onwards to 2025.

## TFR

First we consider not applying a seasonal difference of 12. The variable was first log transformed to remove the heterogeneity. Then a KPSS (Kwiatkowski-Phillips-Schmidt-Shin) test was conducted to check if the data is stationary. **Table 2** shows the test statistics, it's p-value and the number of differences needed. Here it requires a difference of order 2.

Table 2: The KPSS test on Log transformed TFR. Rounded to 3 significant figures. It's p-value is less than 0.05. So reject the null hypothesis that's it's stationary. The number of differences needed is 2.

| kpss_stat | kpss_pvalue | ndiffs |
|-----------|-------------|--------|
| 1.2       | 0.01        | 2      |

In **Figure 3**, the assumption of stationarity is met. The log transformed TFR's ACF and PACF can now be evaluated for a candidate model. With the seasonal ACF, the lags are significant at 12 and 24. There is exponential decay for the seasonal lags here. Whereas on the PACF, the only significant correlating seasonal lag is 12. An appropriate seasonal component for the SARIMA is autoregressive,  $(1, 0, 0)_{12}$ .

For the non-seasonal component, in the ACF, the last significant lag is 1. There is a damped sine-wave decay zero. The PACF however, the last significant lag is at 2 but only lag 1 correlated with the ACF. So concluded an moving average component that's twice differenced;  $(1, 2, 0)$

Hence assuming white noise  $Z_t \sim \text{WN}(0, \sigma^2)$  an initial candidate model in the backward shift difference notation is SARIMA(1, 2, 0)(1, 0, 0)<sub>12</sub>:

$$\phi_1(B)\Phi_1(B^{12})\nabla^2 \log(\text{TFR}_t) = Z_t$$

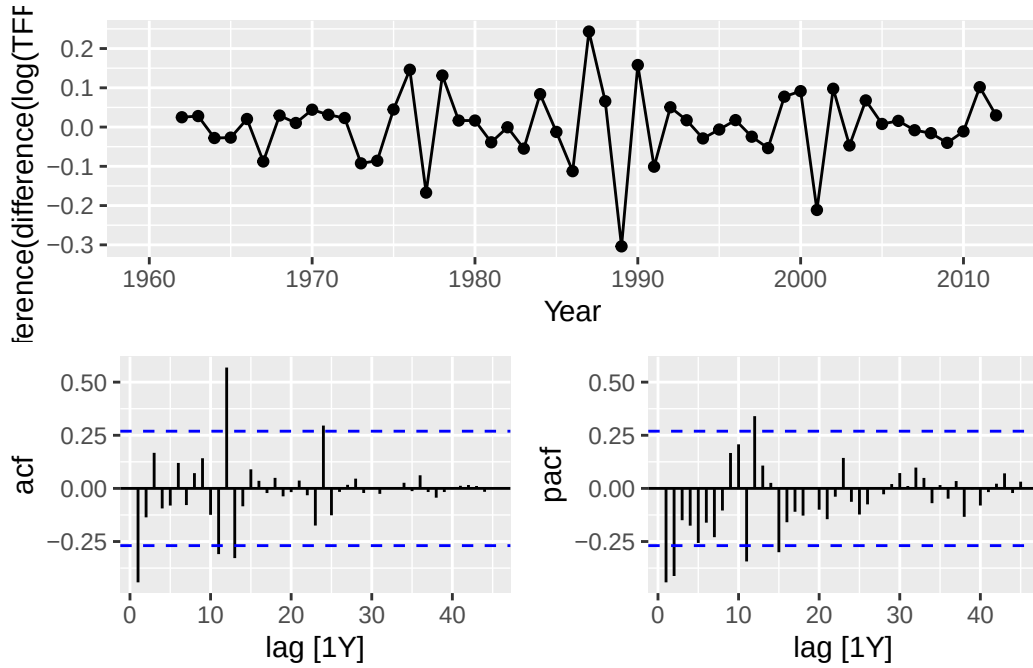


Figure 3: The time plot of the twice differenced log transformed TFR, it's ACF and PACF. The asumption of stationarity is met here. It's wondering around zero and there no obivious patterns

Now a seasonal difference of 12 years is implemented before removing the trend via a first order difference. Using the KPSS statistics again, it's shown that in **Table 3** the null-hypothesis



is rejected as it is not stationary, there is still a first order difference needed to remove the trend.

Table 3: The KPSS test on Log transformed TFR. It's p-value is less than 0.05. So we reject the null hypothesis that's it's stationary. The number of differences needed is 1.

| kpss_stat | kpss_pvalue | ndiffs |
|-----------|-------------|--------|
| 0.639     | 0.0191      | 1      |

**Figure 4** shows that the time plot is now stationary. Only the seasonal lags 12 on the ACF is significant while the PACF displays insignificant seasonal lags that are exponentially decaying. So a auto-regression of 1 for the seasonal component will be considered.

For the non-seasonal lags, on the ACF, it's last significant is 13, however it being too close to the seasonal lags, there could be an effect on it. This is the case at lag 8 as well. So lag 1 is significant on the ACF. Additionally there is a damped sine-wave decay to zero in both correlograms. On the PACF, the last significant lag is 4 with lag 1 also being significant correlating with the ACF plot. Potential models for this would be an AR(1) or AR(4).

Candidate models to be considered, assuming white noise are:

- SARIMA(1, 1, 0)(1, 1, 0)<sub>12</sub>:

$$\phi_1(B)\Phi_1(B^{12})\nabla\nabla_{12}\log(\text{TFR}_t) = Z_t$$

- SARIMA(4, 1, 0)(1, 1, 0)<sub>12</sub>:

$$\phi_4(B)\Phi_1(B^{12})\nabla\nabla_{12}\log(\text{TFR}_t) = Z_t$$

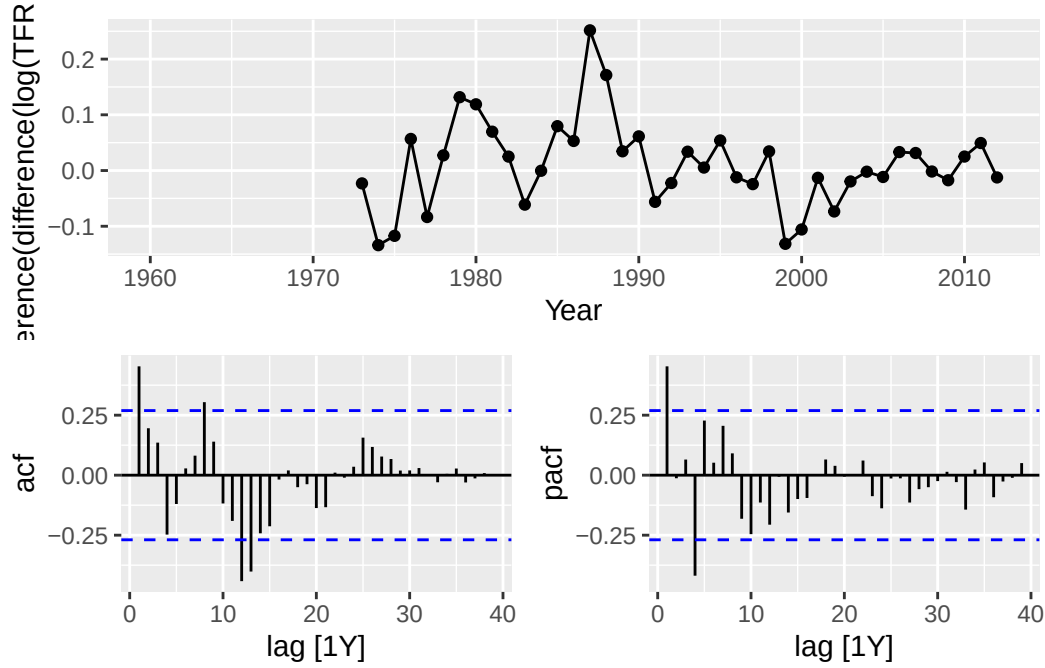


Figure 4: Time plot, ACF and PACF of the log transformed TFR seasonally differenced and first order differenced. It is now stationary, note 12 data points have been removed, leaving only years from 1970s onwards.

The candidate models were fitted to the training data and their AICc evaluated seen in **Table 4**. Amongst the two models with the same total differencing, the AR(1) model has the lowest AICc of -100.

Table 4: Model selection statistics of SARIMA models for the log transforms of TFR based on AICc. Rounded to 3 significant figures

| .model      | sigma2  | log_lik | AIC  | AICc  | BIC   |
|-------------|---------|---------|------|-------|-------|
| arima120100 | 0.00448 | 64.4    | -123 | -122  | -117  |
| arima110110 | 0.00411 | 53.1    | -100 | -99.5 | -95.1 |
| arima410110 | 0.00376 | 56.5    | -101 | -98.4 | -90.8 |

When comparing with the the root mean squared error, the AR(4) model has the lowest value in **Table 5** despite having a slightly higher AICc. The twice first order differenced model has the largest RMSE of 1.18. So the best possible model is an AR(4).

Table 5: Forest Accuracy Statistics of SARIMA models for the log transforms of TFR based on RMSE. Rounded to 3 significant figures

| .model      | .type | ME     | RMSE  | MAE    | MPE   | MAPE |
|-------------|-------|--------|-------|--------|-------|------|
| arima410110 | Test  | 0.0498 | 0.101 | 0.0886 | 3.78  | 7.96 |
| arima110110 | Test  | 0.066  | 0.105 | 0.0952 | 5.3   | 8.45 |
| arima120100 | Test  | -0.84  | 1.18  | 0.84   | -84.8 | 84.8 |

The residuals are then evaluated to check if they meet the white noise assumption for the selected model in **Figure 5**. A Ljung-box test is then use to reconfirm the white noise assumption in **Table 6**.

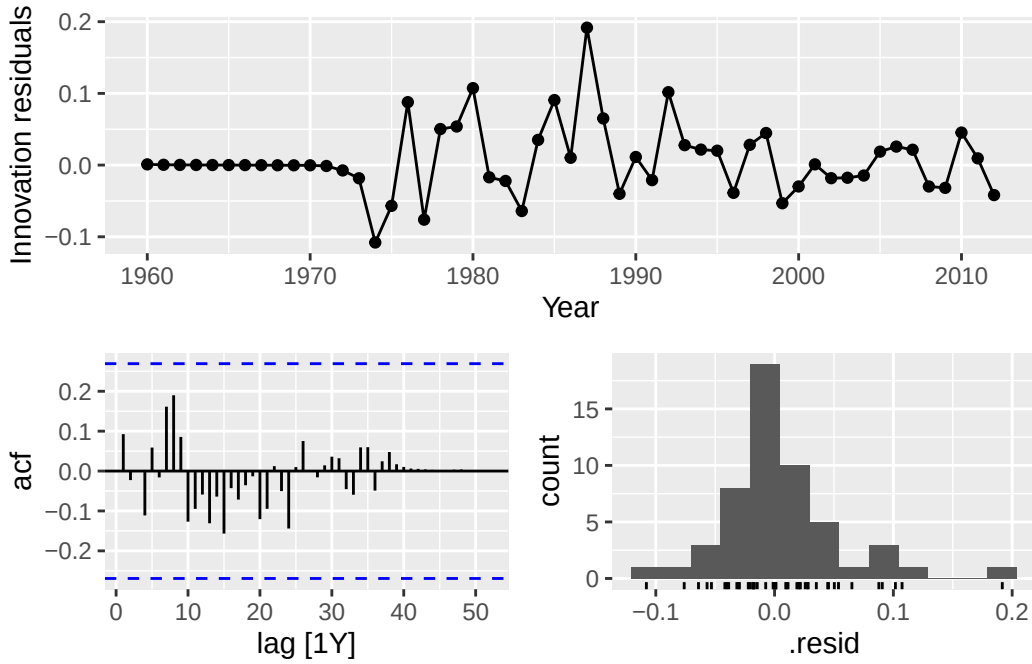


Figure 5: Time plot, ACF and histogram of residuals for selected best model. No significant spikes in the ACF for lags, the residuals are appropriately white noise. The histogram centred around zero with an outlier around 0.2. Therefore can be used for forecasting

Table 6: Ljung-Box Test of the selected AR(4) model. The degree's of freedom are the parameters for the AR and MA components. It's p-value is greater than 0.05 so no significant difference compared to white noise

| .model      | lb_stat | lb_pvalue |
|-------------|---------|-----------|
| arima410110 | 16.5    | 0.627     |

## TFB

Log transformed the total live births to remove the decreasing trend. Then first consider differencing by the 12 year cycle. Differencing again by the first order according to the time plot and KPSS test statistics. The seasonal lags in **Figure 6** for the ACF display an exponential decay, with only lag 12 being significant. Seasonal PACF has no significant seasonal lags but a exponential decay. An AR(1) will be considered.

For the non-seasonal component, in the ACF, the last significant lag is 13 while the PACF the last significant is 10. Both too close to seasonal lags to be considered. So PACF lag 4 is significant and correlates with the ACF at the same lag. Hence an AR(4) model will be considered.

- SARIMA(4, 1, 0)(1, 1, 0)<sub>12</sub>

$$\phi_4(B)\Phi_1(B^{12})\nabla\nabla_{12}\log(\text{TLB}_t) = Z_t$$

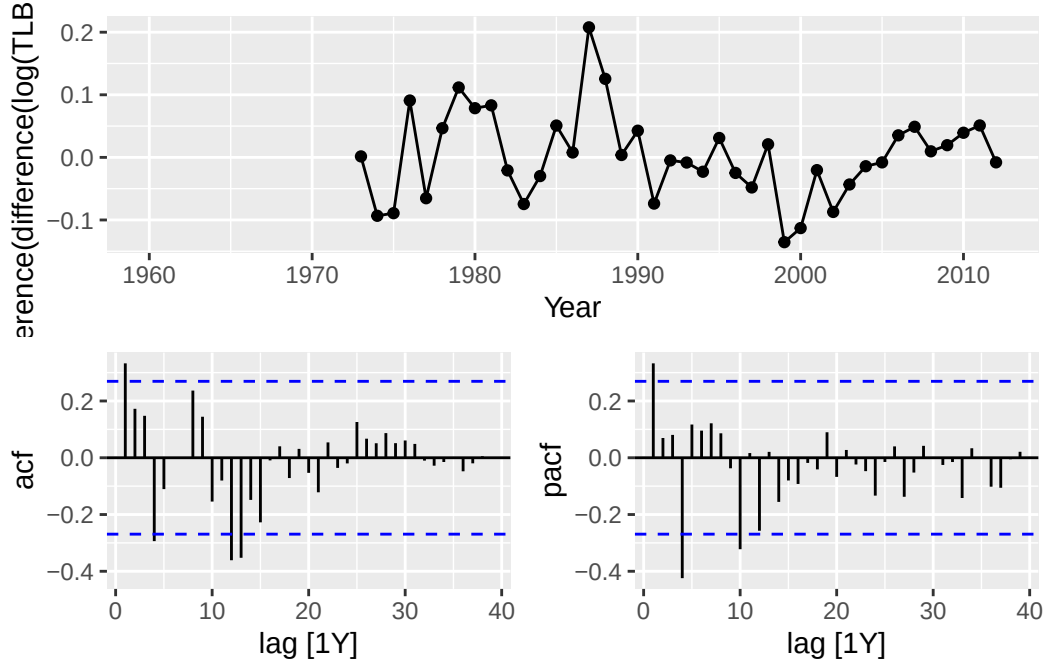


Figure 6: Time plot, ACF and PACF of the log transformed TLB seasonally differenced and first order differenced. It is now stationary, note 12 data points have been removed, leaving only years from 1970s onwards.

Additionally considered only the seasonal difference in **Figure 6**. The KPSS test statistic in **Table 7** after differencing by 12 years shows that the time plot is stationary and no first order differencing is required.

Table 7: The KPSS test on Log transformed TKB. Rounded to 3 significant figures. It's p-value is greater than 0.05. Accept the null hypothesis that's it's stationary. No first order difference needed

| kpss_stat | kpss_pvalue | ndiffs |
|-----------|-------------|--------|
| 0.234     | 0.1         | 0      |

In **Figure 7** Seasonal ACF and PACF lags are difficult to determine, however on the ACF, only lag 12 is significant. Considered a seasonal component of no MA and AR. As well as a an AR(1) model.

The ACF shows a prominent sine wave decay while the PACF is possibly an exponential decay. Last significant lag in the PACF is 5 so considered an AR(5) model.

- SARIMA(5, 0, 0)(0, 1, 0)<sub>12</sub>

$$\phi_5(B)\nabla_{12}\log(\text{TLB}_t) = Z_t$$

- SARIMA(5, 0, 0)(1, 1, 0)<sub>12</sub>

$$\phi_5(B)\Phi_1(B^{12})\nabla_{12}\log(\text{TLB}_t) = Z_t$$

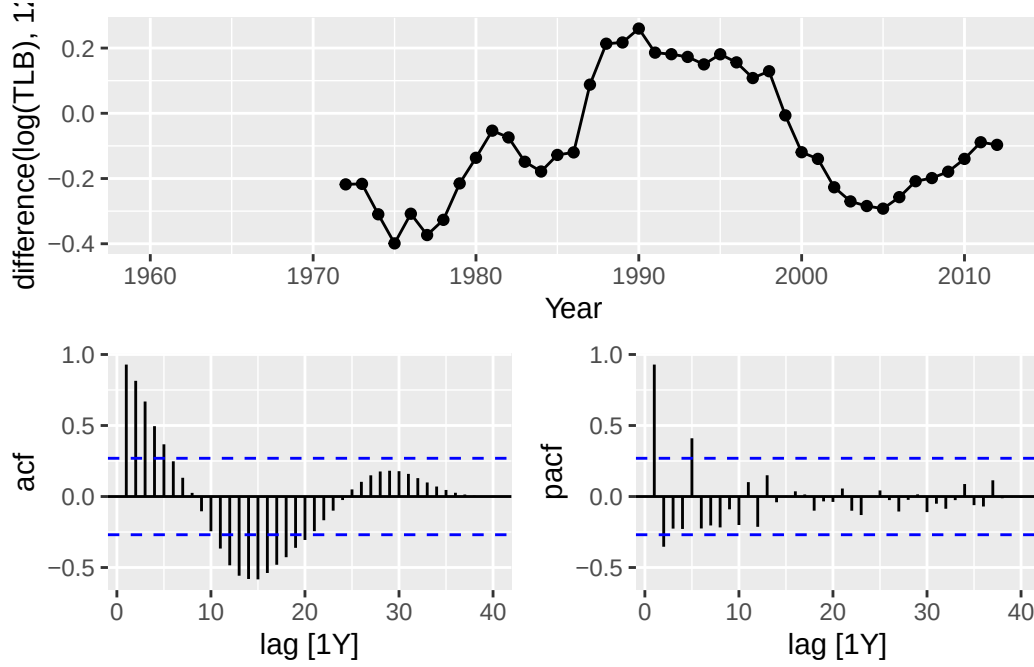


Figure 7: Time plot, ACF and PACF of the log transformed TLB seasonally differenced only. It is stationary according to the KPPS test, note 12 data points have been removed, leaving only years from 1970s onwards. A strong sine decaying sine wave is present in the ACF

Candidate models have been fitted the test data in **Table 8**. Based on AICc alone the best possible model amongst the models with no first order differencing is seasonal MA(1) with an AICc of -105.

Table 8: Model selection statistics of SARIMA models for the log transforms of TLB based on AICc. Rounded to 3 significant figures. Note despite being unable to compared between models with differing numbers of differencing, the AR(4) model has the lowest AICc

| .model      | sigma2  | log_lik | AIC  | AICc | BIC   |
|-------------|---------|---------|------|------|-------|
| arima410011 | 0.00246 | 60.2    | -108 | -106 | -98.2 |
| arima500011 | 0.00278 | 61.1    | -108 | -105 | -96.3 |
| arima500010 | 0.0035  | 58.9    | -106 | -103 | -95.5 |

Further model comparison is done, based on the RMSE in **Table 9**. See that an AR(4) model with both a first order and seasonal difference has the lowest RMSE of 3370. This is an unusually highly positive value and will be further discussed in the **Discussion and Conclusion**. However, out of the manually created candidate models, the selected as the best possible model is this.

Table 9: Forest Accuracy Statistics of SARIMA models for the log transforms of TLB based on RMSE. Rounded to 3 significant figures

| .model      | .type | ME    | RMSE | MAE  | MPE    | MAPE |
|-------------|-------|-------|------|------|--------|------|
| arima410011 | Test  | 555   | 3370 | 2750 | 0.632  | 7.48 |
| arima500011 | Test  | -1410 | 4540 | 3440 | -4.86  | 9.76 |
| arima500010 | Test  | 353   | 4930 | 4010 | -0.308 | 11   |

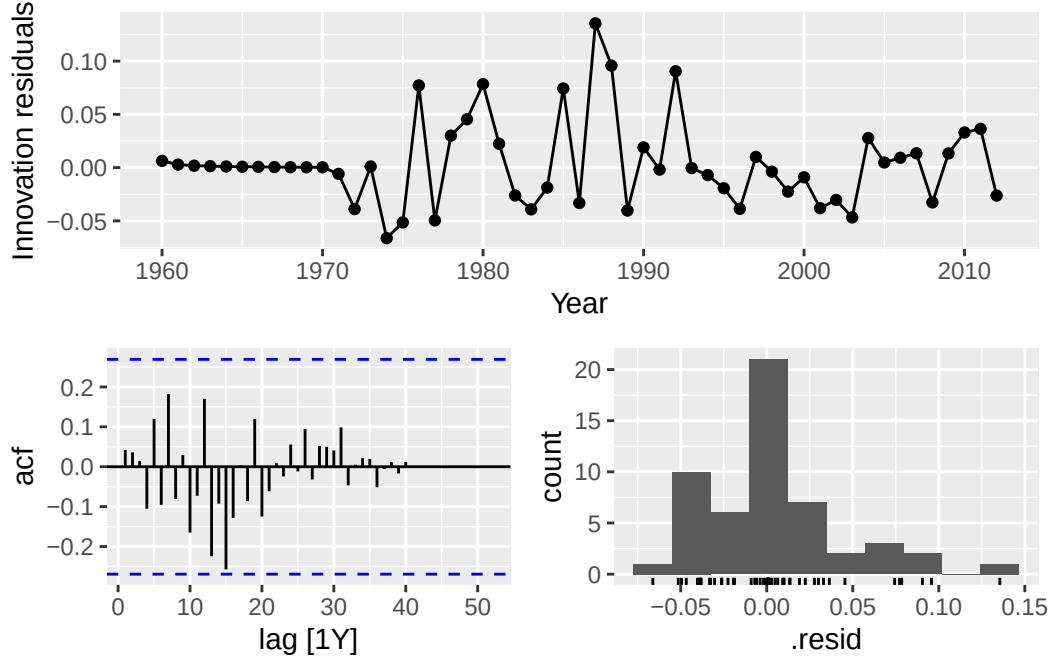


Figure 8: Time plot, ACF and histogram of residuals for selected best model. No significant spikes in the ACF for lags. The residuals are appropriately white noise. The histogram centred around zero with an outlier between 0.10 and 0.15. Therefore can be used for forecasting

Table 10: Further checking with Ljung-Box Test of the selected model. The degree's of freedom are the parameters for the AR and MA components. It's p-value is greater than 0.05 so no significant difference compared to white noise

| .model      | lb_stat | lb_pvalue |
|-------------|---------|-----------|
| arima410011 | 23.8    | 0.205     |

## Results

Candidate models from **Methods** were fitted and their AICc, and MSE, along with their the residuals were evaluated. The best possible model was selected and used to forecast the Test data.

## TFR

SARIMA(4, 1, 0)(1, 1, 0)<sub>12</sub>:



$$\phi_4(B)\Phi_1(B^{12})\nabla\nabla_{12}\log(\text{TFR}) = Z_t$$

- $\phi_4(B) = 1 - 0.450B + 0.059B^2 - 0.333B^3 + 0.405B^4$
- $\Phi_1(B^{12}) = 1 + 0.334B^{12}$

Table 11: Parameter estimates of the selected model for TFR, rounded to 3 significant figures

| term | estimate | std.error | statistic | p.value |
|------|----------|-----------|-----------|---------|
| ar1  | 0.45     | 0.144     | 3.12      | 0.00338 |
| ar2  | -0.0592  | 0.163     | -0.364    | 0.718   |
| ar3  | 0.333    | 0.163     | 2.05      | 0.0474  |
| ar4  | -0.405   | 0.158     | -2.56     | 0.0145  |
| sar1 | -0.334   | 0.166     | -2.01     | 0.0509  |

Table 12: Model fit statistics of the selected model for TFR. Rounded to 3 significant figures

| Statistics | Value        |
|------------|--------------|
| sigma2     | 0.0037551    |
| log_lik    | 56.4889689   |
| AIC        | -100.9779379 |
| AICc       | -98.4324833  |
| BIC        | -90.8446612  |

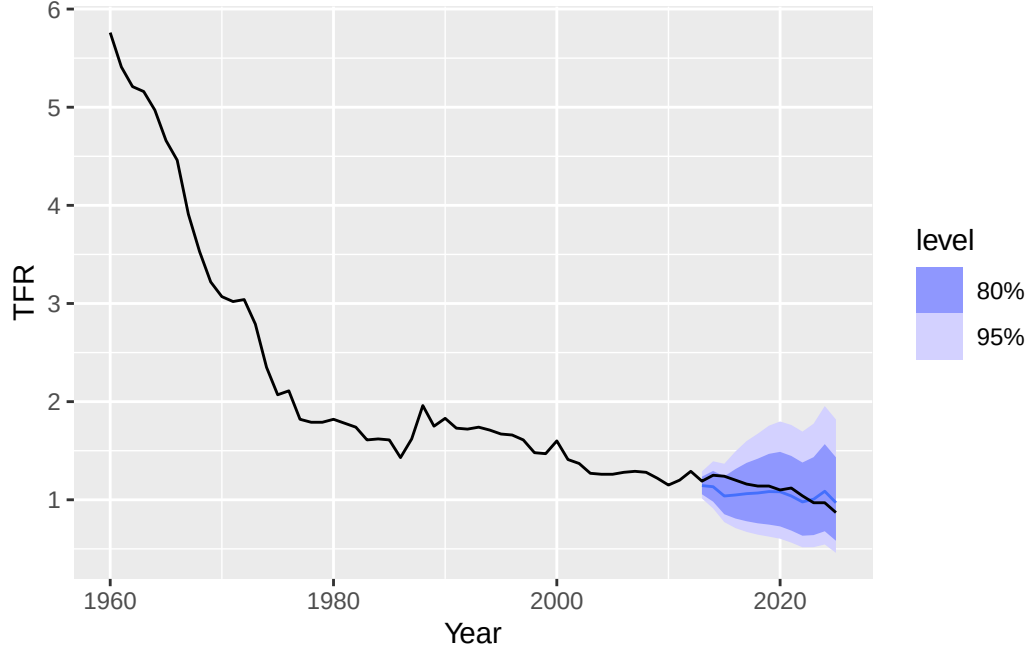


Figure 9: The projected forecast of TFR by the model. See that over time, the projection models the downward trend. Uncertainty widens over time which is within expectation of Time Series. Note the cyclical pattern isn't well captured here by the SARIMA model. As from contextual information, the prominence of this pattern only occurred from 1970s onwards. It's possible that there isn't enough of the data too completely capture this cycle. Overall the projection of the test data is still within the 95% prediction intervals.

## TLB

SARIMA(4, 1, 0)(1, 1, 0)<sub>12</sub> :

$$\phi_4(B)\Phi_1(B^{12})\nabla\nabla_{12}\log(\text{TLB}) = Z_t$$

- $\phi_4(B) = 1 - 0.293B - 0.0507B^2 - 0.243B^3 + 0.444B^4$
- $\Phi_1(B^{12}) = 1 + 0.302B^{12}$

Table 13: Parameter estimates of the selected model for TLB. Rounded to 3 significant figures

| term | estimate | std.error | statistic | p.value |
|------|----------|-----------|-----------|---------|
| ar1  | 0.293    | 0.142     | 2.07      | 0.0452  |
| ar2  | 0.0507   | 0.156     | 0.325     | 0.747   |

| term | estimate | std.error | statistic | p.value |
|------|----------|-----------|-----------|---------|
| ar3  | 0.243    | 0.148     | 1.65      | 0.108   |
| ar4  | -0.444   | 0.15      | -2.97     | 0.005   |
| sar1 | -0.302   | 0.169     | -1.79     | 0.0818  |

Table 14: Model fit statistics of the selected model for TLB. Rounded to 3 significant figures

| Statistics | Value   |
|------------|---------|
| sigma2     | 0.00334 |
| log_lik    | 59.2    |
| AIC        | -106    |
| AICc       | -104    |
| BIC        | -96.2   |

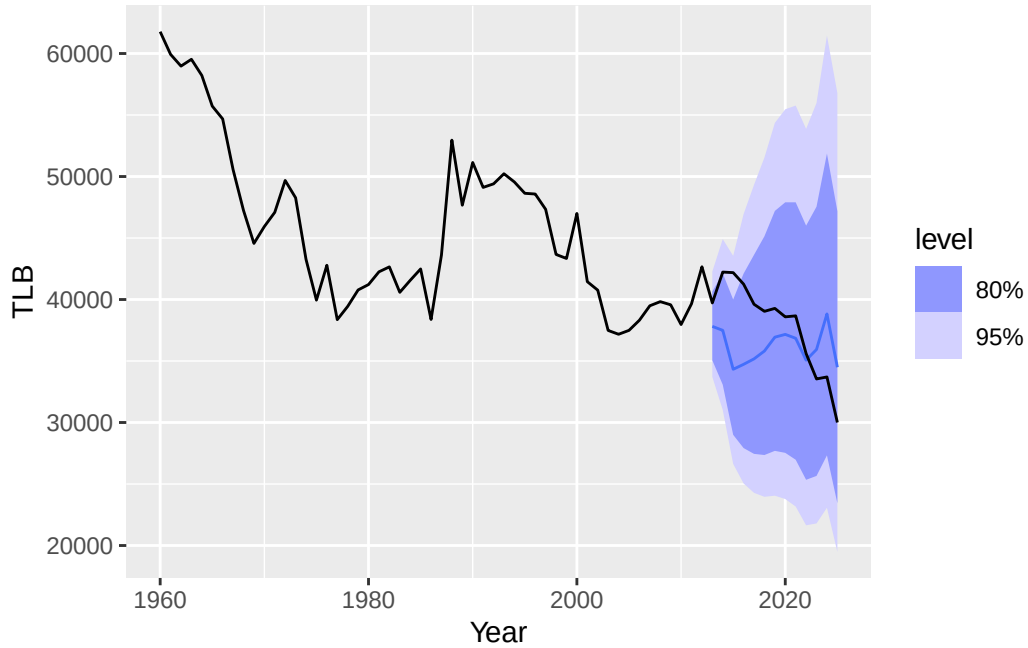


Figure 10: The projected forecast of TLB by the selected model. This has wider prediction bands suggesting higher volatility and fluctuations. Fails to capture the decreasing trend and cyclical nature of the data. Hence a SARIMA model may not be best at modelling total live births despite still being within the 95% prediction intervals.

## References

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## Statistical Appendix

- A full mathematical description of any time series model, statistical test and estimation procedure.
- [Github Repository](#)

### SARIMA Models $\text{SARIMA}(p, d, q)(P, D, Q)_m$

For seasonal ARIMA with  $m$  periods with white noise assumption, the general model is:

$$\phi(B)\Phi(B^m)\nabla^d\nabla_m^D X_t = \theta(B)\Theta(B^m)Z_t$$

- The AR(p) and Seasonal AR(P)

$$\phi(z) = 1 - \sum_{i=1}^p \phi_i z^i$$

$$\Phi(z) = 1 - \sum_{i=1}^P \Phi_i z^i$$

- The  $d$  order Difference and Seasonal Difference  $D$

$$\nabla^d = (1 - B)^d$$

$$\nabla_m^D = (1 - B^m)^D$$

- The MA(q) and Seasonal MA(Q)

$$\theta(z) = 1 - \sum_{i=1}^q \theta_i z^i$$

$$\Theta(z) = 1 - \sum_{i=1}^Q \Theta_i z^i$$

### Autocorrelation

For lag  $k$ , the ACF is the measure between the relationship between  $y_t$  and  $y_{t-k}$ :

$$\rho(k) = r_k = \frac{\sum_{t=k+1}^T (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2}$$

### Partial Autocorrelation

The theoretical partial autocorrelation function of a stationary time series is calculated using the Durbin-Levinson Algorithm

$$\phi_{n,n} = \frac{\rho(n) - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(n-k)}{1 - \sum_{k=1}^{n-1} \phi_{n-1,k} \rho(k)}$$

Where

- $\phi_{n,k} = \phi_{n-1,k} - \phi_{n,n} \phi_{n-1,n-k}$  for  $k \in [1, n-1]$
- $\rho(\cdot)$  is the autocorrelation function

### KPSS Test

- The null-hypothesis is Stationary. Reject if is greater than 0.05

### Ljung-Box Test

- $H_0$ : The data is not correlated (white noise assumption)
- $H_a$ : The data exhibits serial correlation
- Test statistics

$$Q = n(n+2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{n-k}$$

Where  $n$  is the sample size  $\hat{\rho}_k$  us the sample autocorrelation at lag  $k$  and  $h$  is the number of lags tested.

Under  $H_0$ ,  $Q$  asymptotically follows  $\chi_h^2$

So for significance level  $\alpha$ , reject  $H_0$  if

$$Q > \chi_{1-\alpha,h}^2$$

or if P-value is greater than 0.05 reject the  $H_0$

**Code used in the Report:**

```

# Load library
pacman::p_load(fpp3)
# Load the cleaned Dataset
read.csv(
  here::here(
    "AnnualFertilityData_Cleaned.csv"), # This can be found on github
  check.names = FALSE
) |>
  tsibble(index = Year) -> fert_df

zodiac <- c( # Set up the Zodiac Cycle and bind it to the data
  "Rat", "Ox", "Tiger", "Rabbit",
  "Dragon", "Snake", "Horse", "Goat",
  "Monkey", "Rooster", "Dog", "Pig"
)

fert_df |>
  mutate(Zodiac = zodiac[((Year - 1900) %% 12 + 1)]) |>
  relocate(Zodiac, .after = Year) -> fert_df

# Train and Split datasets
train <-
  fert_df |>
  select(Year, TFR, TLB, Zodiac) |>
  filter(Year <= 2012)

test <-
  fert_df |>
  select(Year, TFR, TLB, Zodiac) |>
  filter(Year > 2012)

# Model Fitting on Train
TFR_fit <-
  train |>
  model(
    arima120100 = ARIMA(log(TFR) ~ pdq(1,2,0) + PDQ(1,0,0, period = 12)),
    arima110110 = ARIMA(log(TFR) ~ pdq(1,1,0) + PDQ(1,1,0, period = 12)),
    arima410110 = ARIMA(log(TFR) ~ pdq(4,1,0) + PDQ(1,1,0, period = 12)))

TLB_fit <-
  train |>
  model(
    arima410011 = ARIMA(log(TLB) ~ pdq(4,1,0) + PDQ(0,1,1, period = 12)),

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    arima500010 = ARIMA(log(TLB) ~ pdq(5,0,0) + PDQ(0,1,0, period = 12)),
    arima500011 = ARIMA(log(TLB) ~ pdq(5,0,0) + PDQ(0,1,1, period = 12)))
# Model Fitting on Test
TFR_bestFit <-
  train |>
  model(arima410110 = ARIMA(log(TFR) ~ pdq(4,1,0) + PDQ(1,1,0, period = 12)))

TLB_bestFit <-
  train |>
  model(arima410110 = ARIMA(log(TLB) ~ pdq(4,1,0) + PDQ(1,1,0, period = 12)))
# Table 1
fert_df |>
  select(Year, Zodiac) |>
  tail(12) |>
  knitr::kable(
    caption =
      "The last twelve years and their corresponding Zodiac Animal")
# Fig1
fert_df |>
  ggplot(aes(x = Year, y = TFR, label = Year, )) +
  geom_line() +
  geom_point(colour = "gray") +
  geom_point(aes(colour = Zodiac),
    data = filter(fert_df, Zodiac %in% c("Dragon", "Tiger"))) +
  scale_x_continuous(breaks = filter(fert_df,
    Zodiac %in% c("Dragon", "Tiger"))$Year) +
  ggsci::scale_color_lancet("lanonc") +
  theme(axis.text.x = element_text(angle = 50, hjust = 1),
    legend.position = 'bottom')
# Fig2
fert_df |>
  ggplot(aes(x = Year, y = TLB, label = Year, )) +
  geom_line() +
  geom_point(colour = "gray") +
  geom_point(aes(colour = Zodiac),
    data = filter(fert_df, Zodiac %in% c("Dragon", "Tiger"))) +
  scale_x_continuous(breaks = filter(fert_df,
    Zodiac %in% c("Dragon", "Tiger"))$Year) +
  ggsci::scale_color_lancet("lanonc") +
  theme(axis.text.x = element_text(angle = 50, hjust = 1),
    legend.position = 'bottom')
#Tab2

```



```

train |>
  features(log(TFR), list(unitroot_kpss,
                          unitroot_ndiffs)) |>
mutate_if(is.numeric, ~ as.character(signif(.,3))) |>
knitr::kable(
  caption =
    "The KPSS test on Log transformed TFR.
    Rounded to 3 significant figures.
    It's p-value is less than 0.05.
    So reject the null hypothesis that's it's stationary.
    The number of differences needed is 2." )

#Fig3
train |>
  gg_tsdisplay(log(TFR) |>
    difference(1) |>
    difference(1),
    plot_type = 'partial', lag_max = 45)

#Tab3
train |>
  features(log(TFR) |>
    difference(12), list(unitroot_kpss,
                          unitroot_ndiffs)) |>
mutate_if(is.numeric, ~ as.character(signif(.,3))) |>
knitr::kable(
  caption =
    "The KPSS test on Log transformed TFR.
    It's p-value is less than 0.05.
    So we reject the null hypothesis that's it's stationary.
    The number of differences needed is 1." )

#Fig4
train |>
  gg_tsdisplay(log(TFR) |>
    difference(12) |>
    difference(1),
    plot_type = 'partial', lag_max = 45)

#Tab4
glance(TFR_fit) |>
  arrange(AICc) |>
  select(.model:BIC) |>
mutate_if(is.numeric, ~ as.character(signif(.,3))) |>
knitr::kable(

```

```

caption =
  "Model selection statistics of
  SARIMA models for the log transforms of TFR based on AICc.
  Rounded to 3 significant figures")
#Tab5
TFR_fit |>
  forecast(new_data = test) |>
  accuracy(test) |>
  arrange(RMSE) |>
  select(.model:MAPE) |>
  mutate_if(is.numeric, ~ as.character(signif(.,3))) |>
  knitr::kable(
    caption =
      "Forest Accuracy Statistics of
      SARIMA models for the log transforms of TFR based on RMSE.
      Rounded to 3 significant figures")

#Fig5
TFR_fit |>
  select(arima410110) |>
  gg_tsresiduals(lag = 60)
#Tab6
augment(TFR_fit) |>
  filter(.model == "arima410110") |>
  features(.innov, ljung_box, lag = 24, dof = 5) |>
  mutate_if(is.numeric, ~ as.character(signif(.,3))) |>
  knitr::kable(
    caption =
      "Ljung-Box Test of the selected AR(4) model.
      The degree's of freedom are the
      parameters for the AR and MA components.
      It's p-value is greater than 0.05 so
      no significant difference compared to white noise")
#Fig6
train |>
  gg_tsdisplay(log(TLB) |>
    difference(12) |>
    difference(1),
    plot_type = 'partial', lag_max = 45)
#Tab7
train |>
  features(log(TLB) |>

```

```

        difference(12), list(unitroot_kpss,
                             unitroot_ndiffs)) |>
mutate_if(is.numeric, ~ as.character(signif(.,3))) |>
knitr::kable(
  caption =
    "The KPSS test on Log transformed TKB.
    Rounded to 3 significant figures.
    It's p-value is greater than 0.05.
    Accept the null hypothesis that's it's stationary.
    No first order difference needed")
#Fig7
train |>
  gg_tsdisplay(log(TLB) |>
    difference(12), plot_type = 'partial', lag_max = 45)
#Tab8
glance(TLB_fit) |>
  arrange(AICc) |>
  select(.model:BIC) |>
  mutate_if(is.numeric, ~ as.character(signif(.,3))) |>
  knitr::kable(
    caption =
      "Model selection statistics of
      SARIMA models for the log transforms of TLB based on AICc.
      Rounded to 3 significant figures.
      Note despite being unable to
      compared between models
      with differing numbers of differencing,
      the AR(4) model has the lowest AICc" )
#Tab9
TLB_fit |>
  forecast(new_data = test) |>
  accuracy(test) |>
  arrange(RMSE) |>
  select(.model:MAPE) |>
  mutate_if(is.numeric, ~ as.character(signif(.,3))) |>
  knitr::kable(
    caption =
      "Forest Accuracy Statistics of
      SARIMA models for the log transforms of TLB based on RMSE.
      Rounded to 3 significant figures")
#Fig8
TLB_fit |>

```

```

select(arima410011) |>
gg_tsresiduals(lag = 60)
#Tab10
augment(TLB_fit) |>
filter(.model == "arima410011") |>
features(.innov, ljung_box, lag = 24, dof = 5) |>
mutate_if(is.numeric, ~ as.character(signif(.,3))) |>
knitr::kable(
  caption =
    "Further checking with Ljung-Box Test of the selected model.
    The degree's of freedom are the parameters
    for the AR and MA components.
    It's p-value is greater than 0.05
    so no significant difference compared to white noise")
#Tab11
TFR_bestFit |>
tidy() |>
select(term:p.value) |>
mutate_if(is.numeric, ~ as.character(signif(.,3))) |>
knitr::kable(
  caption =
    "Parameter estimates of
    the selected model for TFR, rounded to 3 significant figures")
#Tab12
TFR_bestFit |>
glance() |>
select(sigma2:BIC) |>
pivot_longer(
  cols = everything(),
  names_to = "Statistics",
  values_to = "Value"
) |>

knitr::kable(
  caption =
    "Model fit statistics of the selected model for TFR.
    Rounded to 3 significant figures")
#Fig9
TFR_bestFit |>
forecast(new_data = test) |>
autoplot(fert_df)
TLB_bestFit |>

```

```

tidy() |>
select(term:p.value) |>
mutate_if(is.numeric, ~ as.character(signif(.,3))) |>
knitr::kable(
  caption =
    "Parameter estimates of the selected model for TLB.
    Rounded to 3 significant figures")
#Tab14
TLB_bestFit |>
glance() |>
select(sigma2:BIC) |>
pivot_longer(
  cols = everything(),
  names_to = "Statistics",
  values_to = "Value"
) |>
mutate_if(is.numeric, ~ as.character(signif(.,3))) |>
knitr::kable(
  caption =
    "Model fit statistics of the selected model for TLB.
    Rounded to 3 significant figures")
#Fig10
TLB_bestFit |>
forecast(new_data = test) |>
autoplot(fert_df)

```